
Channels to Targets: A Closed-Form Coordinate System for Silence-Blindness in LLM Observers

Ali Khan

Rutgers University

abk135@rutgers.edu

Code: <https://github.com/akhan531/Control-On-LLMs>

Abstract

Large language models are often evaluated on their ability to reason from evidence that is presented in their input. We study a complementary question: can a model reason from evidence that is not explicitly written, but is derivable from a known disclosure rule? We introduce a controlled Bayesian framework for quantifying this ability. The framework holds the underlying posterior fixed while varying how evidence is represented: explicitly, through an enumerated record of absent observations, or implicitly, where the model must derive those absences from the disclosure rule. This separates the ability to identify implicit evidence from the ability to incorporate it into a belief. Across five model configurations that demonstrate a baseline inferential ability, we find a clear asymmetry: among 150 draws, 103 reach the target belief, and 47 miss it, with all 47 errors under-weighting the evidence conveyed by silence rather than over-weighting. Furthermore, we find that explicitly enumerating absences substantially reduces the tendency for the observer to hold the silence-blind belief, but does not consistently recover the full Bayesian update. These results suggest that even when the relevant evidence is logically recoverable from the transcript, identifying and interpreting that evidence can be distinct failure modes.

1 Introduction

An observer reads a transcript where N analysts each ran independent assays to determine which of two candidates is responsible for a contamination event. From all the information given in the transcript, the observer must output the name of the candidate it thinks is responsible and how confident it is in its answer. However, the information the transcript shows can vary, like if only k of the analysts report the result of their assay, while the remaining $N - k$ report nothing. The reports that are never filed are mentioned in the transcript, and whether the observer reads a missing report as a finding, as noise, or as nothing at all impacts the belief it should hold, so an observer that conditions only on the filed reports may miss out on key evidence.

This situation is small enough to solve exactly through Bayesian inference. Once the specific features of information, or channels, a transcript exposes are written down, the target posterior the observer should hold and the operations it must perform to reach that target both follow in closed form. Therefore, an elicited answer is not just right or wrong, but exists on an interval with reference beliefs. Furthermore, the distance of the observer’s answer to these reference beliefs are measurable.

The framework’s asset is that it maps channels in the transcript to target posteriors, and it is many to one: transcripts with different channels require the observer to perform different operations to absorb all the evidence available. However, if the target posterior is the same, a comparison across them holds the target fixed and varies only what the observer must do to reach it. This framework

gives us a mathematical coordinate system to benchmark an LLM’s ability to reason with evidence that is not explicitly written but is derivable from a disclosure rule.

- **A framework.** A closed-form map from the channels a transcript exposes to the operations required and the target posterior the observer should reach, with two reference beliefs and three conditions that share one target under different required operations.
- **Localizing the direction of error.** Among the five configurations admitted by the competence gate, in the two conditions that do not enumerate the absences, under-weighting is the only direction of error observed, and a configuration resides on the silence-blind belief more often when the absences must be identified than when the reports are printed.
- **Separating identification from interpretation.** Enumerating the absences repairs a configuration’s failure to notice them without repairing its failure to understand them.
- **The instrument’s own limits, reported as results.** Principled design choices that limit the freedom and consistency of experiments that are able to be ran.

2 The framework

Two candidates $A = \{a_0, a_1\}$ are under consideration and exactly one is responsible, with a uniform prior. We take $|A| = 2$, and this one-dimensionality makes this framework truly a "coordinate system", and it is the source of a structural limitation mentioned in Section 6. Each of the N analysts runs a binary assay returning POSITIVE or NEGATIVE. If a_0 is responsible, an assay returns POSITIVE with probability p ; if a_1 is responsible, an assay returns POSITIVE with probability $1 - p$. The results are independent given which candidate is responsible. We take $1/2 < p < 1$, so a POSITIVE is more consistent with a_0 than with a_1 and it tilts the reference belief toward a_0 . What the observer’s transcript shows is controlled by a *disclosure rule*, and it is this rule that makes our conditions vary: some display every result, while others only file the k POSITIVE reports and leave the $N - k$ NEGATIVE reports as absences to be recovered through the value of N and the count of filed reports. This is the bookbag-and-poker-chip paradigm of Phillips and Edwards [1966] conditioned on a disclosure rule. Therefore, the transcript displays different channels across different conditions, as seen in Table 1. It is the channel set a transcript exposes that determines both the operations a correct observer must perform and the belief it should hold.

Write $L(a)$ for the probability that a given analyst’s assay returns positive under candidate a , and let k be the number of positives from the N assays. Two beliefs are definable from the transcript alone:

$$b_{\text{pos}}(a) \propto L(a)^k, \quad (1)$$

$$b_{\text{tar}}(a) \propto L(a)^k (1 - L(a))^{N-k}. \quad (2)$$

Equation (1) reads the positive filed reports as if they were the whole record; call it the silence-blind belief b_{pos} . Equation (2) instead scores every absence as a negative finding. Under the disclosure rule, this is the correct posterior; call it the target b_{tar} . These are the two reference beliefs on the interval on which the observer’s answer is located.

This design’s asset is that the conditions that print the negatives, enumerate the absences, or leave them to be derived all encode the same information about the N reports. They all return the same target, which makes their comparison valid.

Lemma 1 (Ordering). Let $|A| = 2$ and $1/2 < p < 1$, and consider a transcript with at least one absence ($N - k \geq 1$). Then the silence-blind belief places strictly higher log-odds on a_0 than the target does: reading the silence moves belief strictly below the silence-blind level.

Proof. Let a_0 be the candidate with $L(a_0) = p$ and a_1 the other, so $L(a_1) = 1 - p$ and $p > 1/2$ gives $L(a_0) > L(a_1)$. Write $\Lambda(b) := \log \frac{b(a_0)}{b(a_1)}$ for the log-odds a belief b places on a_0 . The two beliefs share the presence term $k \log \frac{p}{1-p}$ and differ only in how the $N - k$ absences are scored:

$$\Lambda(b_{\text{pos}}) = k \log \frac{p}{1-p}, \quad \Lambda(b_{\text{tar}}) = k \log \frac{p}{1-p} + (N - k) \log \frac{1-p}{p}.$$

Because $p > 1 - p$, $\log \frac{1-p}{p} < 0$, so with at least one absence ($N - k \geq 1$) every absence strictly lowers the log-odds on a_0 , so $\Lambda(b_{\text{tar}}) < \Lambda(b_{\text{pos}})$. $\square$

Table 1: The map. Each condition’s transcript exposes a different channel set, which determines both the operations a correct observer must perform and the belief it should hold. Rows 2 through 4 differ in operations yet agree on target.

condition	channels in transcript	operations required	target
BLIND	positives	aggregate	b_{pos}
FULL	positives, printed negatives	aggregate	b_{tar}
ARITH	positives, enumerated absences	convert, aggregate	b_{tar}
RULED	positives, derivable absences	identify, convert, aggregate	b_{tar}

Three operations appear in Table 1. To *identify* is to determine if reports are absent and how many; if N is present in the transcript, this boils down to a single subtraction, so noticing the absences and counting them are two different parts of one step. To *convert* is to read an absence as a negative finding. To *aggregate* is to combine the findings into a posterior. The three conditions sharing a target have the nested required operation sets $\{a\} \subset \{c, a\} \subset \{i, c, a\}$, which Example 2 investigates. One honest limit follows: aggregation is never isolated, because no condition in our design sits below $\{a\}$.

The object we measure is the observer’s ability to reason with evidence that is not explicitly written in the transcript but is derivable from understanding its disclosure rule. The reference beliefs give points to compare the observer’s response to: in a condition where the absences are derivable, occupancy of b_{pos} is diagnostic of failing to incorporate the silent evidence, while occupancy of b_{tar} indicates success in using that evidence to reach the target posterior.

3 The instrument

The framework from Section 2 provides the interval with reference beliefs for any observer that reads such a transcript. We now put real LLM observers on it, running the conditions of Table 1 as prompts and plotting each model’s response on the same coordinate. The seven configurations we initially tested span four model families: openai/gpt-5.6-sol, z-ai/glm-5.2, deepseek/deepseek-v4-flash-0731, and meta-llama/llama-3.3-70b-instruct. Every family except the Llama one contributes two configurations that differ in reasoning setting and the paired token budget, which is what the "-high" suffix on a configuration name denotes (abbreviated "-hi" in Figure 1 and Figure 2); Appendix D gives the full table.

3.1 The response scale

Answers are elicited on two axes, the candidate they think is responsible and whether they are "Somewhat confident" or "Very confident" in their answer, giving four ordinal levels. We elicit an ordinal level rather than a raw probability on purpose: a numerical posterior produced by an LLM would confound the ability to reason from the evidence with the ability to carry out the arithmetic of Bayesian updating, so coarsening the answer into four ordinal levels strips out most of the noise of that arithmetic and keeps the signal we really mean to measure, whether the observer moves its belief in the right direction and by how much.

The probability ranges these four levels occupy were measured, not assumed. In a calibration pre-test, each configuration is given the stated probability for a candidate being responsible. With no inference to perform, the LLM responds with its internal mapping from a numerical probability to either "Very confident" or "Somewhat confident" (Appendix B). For six of the seven configurations, the switch that separates the two ordinal levels is at 0.20 on the low side and 0.80 on the high side, with llama being the lone exception. A model is very confident in a candidate once its probability clears 0.80, very confident in the other candidate below 0.20, and merely somewhat confident closer to even odds.

The four levels sit in measured bands with deliberate gaps between them: level 0 is $[0, 0.20]$, level 1 is $[0.40, 0.45]$, level 2 is $[0.55, 0.70]$, and level 3 is $[0.80, 1]$ (Appendix B). Every cell is built so that both the silence-blind belief and the target land inside a band rather than one of these gaps, giving each a definite level to score against.

Four constraints are enforced at build time. Every cell is designed so that the silence-blind belief and the target fall in different levels, which holds in all fourteen cells. A build-time forbidden-word list bans silence vocabulary together with Bayesian and computational vocabulary, so no cues are given in the prompt for what operation it is testing. Every stimulus is tested with both label orderings. Count-control cells invert the correct level, so a configuration that always answers high-confidence is separable from a competent one. These controls are intended to distinguish evidence-sensitive inference from heuristics based on the number of positive reports. Appendix C reproduces the prompts, the response schema, and one rendered stimulus per condition.

3.2 The competence gate

Before reading any configuration against the coordinate, we check that it is able to read the record at all. A competence gate is built on two conditions from the same underlying record. In the printed-negatives condition (FULL), every result, positive and negative, is displayed and completeness is asserted, so reaching the correct answer is a matter of aggregation, while in the derivable-absences condition (RULED), only the positive reports appear, and the negatives must be inferred from the stated analyst count and filing rule. A configuration that fails the printed-negatives condition is excluded, because a configuration that fails both conditions is a floor effect rather than a silence result.

To measure where a model lands on the coordinate, we need a distance in levels. We now define the statistics that report it. We report $s_1 = |\ell_{\text{obs}} - \ell_{\text{tar}}|$ and the signed $s_2 = \ell_{\text{obs}} - \ell_{\text{tar}}$, with positive values meaning under-weighting toward the target, where ℓ_{obs} is the level of observer reports and ℓ_{tar} is the level of the target read off Table 1. With this, five of seven configurations are admitted, at mean s_1 on the printed-negatives condition of 0.200, 0.421, 0.457, 0.752 and 0.764 levels; the two excluded sit at 1.609 (deepseek) and 1.800 (llama). A mean s_1 above one means that, on average, the answer lands more than a full level from the target, so the gate sets these two aside. All results in the next two sections are therefore reported only for those five: the configurations that demonstrate a baseline ability to perform inference with all evidence printed. The results below should then be read as characterizing silence-reading behavior among models with this baseline inference ability and not all LLMs generally.

The campaign holds 5,808 usable draws of 5,950 attempted, at five draws per stimulus; the pre-test described above contributes a further 659 of 672 on a separate stimulus set at three draws per stimulus.

4 Example 1: which direction does the error run?

Whether language models update too little or too much is contested, and the two closest anchors disagree, on the populations that do not overlap. Imran et al. [2025] report that pre-trained-only checkpoints update less than a correct Bayesian update would, with the shortfall shrinking as models scale, measured over token log-probabilities. They describe the shortfall as unexplained and note that it may be a result of how improbable their evidence strings are, so we take the direction from them and not a mechanism. Samanta et al. [2026] evaluate instruction-tuned models across turns and report that smaller models stay near the middle of the interval, while larger ones can push toward the extremes when the evidence is skewed. Under-use relative to Bayes has been called conservatism since Edwards [1968]. Imran et al.’s expected update is elicited from the model, read from its own token probabilities, and where a closed-form reference is available elsewhere, it is computed from evidence the observer can see. Gupta et al. [2025] show that updates track the Bayesian posterior closely when enough of that evidence has accumulated in the model’s context. Our target is computed from the disclosure rule, so the question here is not whether a belief moves correctly on what the transcript displays but whether it moves on what it withholds.

The comparison uses two channel sets, printed negatives against derivable absences, which require different operations but share a target. Because the target lies below the silence-blind belief in every cell, an answer that under-weights the absence of a report stays high, above the target, while one that over-weights it falls below. The first eleven cells have a target on level 0 (Appendix A), so nothing lies below it and over-weighting is structurally inexpressible. We therefore designed three cells, D1–D3, with the target one level up specifically so that an answer can fall on either side of

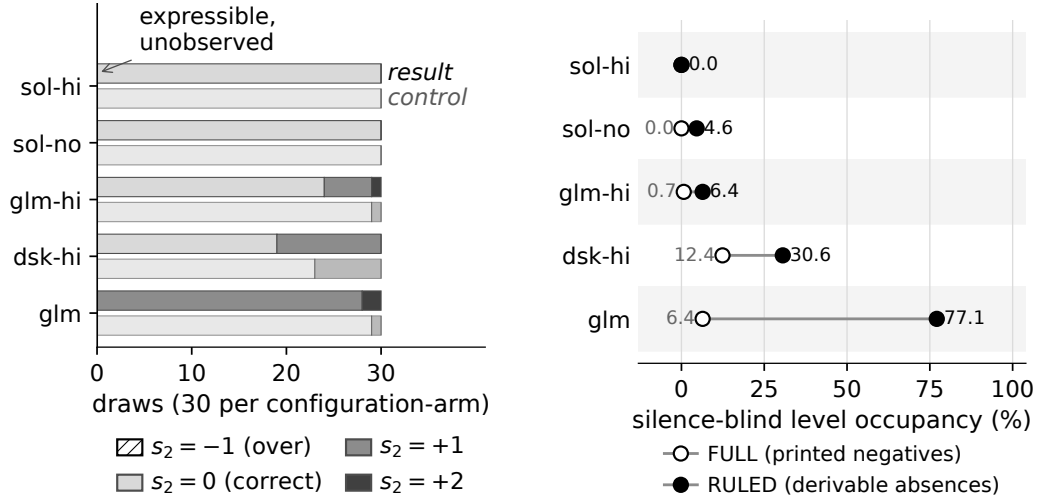


Figure 1: (a) Signed displacement s_2 on the three cells where over-weighting is expressible, five admitted configurations, 150 attempted draws per arm; the result arm is derivable-absences, the control (gate-selected) is printed-negatives. The $s_2 = -1$ (over-weighted) segment has zero width in every bar. (b) Occupancy of the silence-blind level, the percent of draws landing exactly on it, under the printed-negatives condition (open marker) and the derivable-absences one (filled), for the five admitted configurations over all 14 cells. Occupancy rises in four of the five and falls in none, reaching 77.1 percent for glm; sol-hi sits at zero in both. Configurations are abbreviated (dsk=deepseek).

it. These three cells are where under-weighting and over-weighting are both expressible, and the direction-of-error result is measured on them.

The error runs one way. Out of 150 draws spread across the three cells and five admitted configurations, 47 missed the target posterior, and every one of those 47 errors under-weights the silence while not a single draw over-weights it (Figure 1a). The competence gate admits configurations by their performance on the printed-negatives condition and not on the derivable-absences one, so the arm that carries the result played no part in the selection. At the unit of analysis, a configuration's modal answer on a cell, eight of the fifteen configuration-by-cell units are displaced toward under-weighting and none toward over-weighting (two-sided exact sign test, $p = 0.008$); the remaining seven carry no displaced mass. This direction persists in the printed-negatives condition: 9 to 0 on the same 150 draws, five of fifteen units displaced, $p = 0.063$. Among errors that the instrument can express in these cells, the admitted configurations display a strong directional tendency to under-weight the silence rather than over-weight it.

Occupancy. We also ask how often belief resides on the silence-blind belief, and whether that changes based on the channels in the transcript. Occupancy of that level, the fraction of draws that land on it, is measured for the five admitted configurations over all cells, under both the printed-negatives and derivable-absences conditions (Figure 1b). Occupancy of the silence-blind level either rises or stays the same when the negatives must be inferred rather than read, so the evaluated configurations settle on the silence-blind belief more often when the silence has to be recovered.

An ordinal extremity effect predicts mass piling at the endpoints. However, of the 47 displaced draws, 44 land at the level of the silence-blind belief, which is not an endpoint, 3 land at the top level, and none at the bottom, so total endpoint mass is 3 in 150 draws. In all configurations with displaced mass to test, mass at the internal silence-blind level outnumbers mass at the endpoint.

The one-sidedness here is a property of the information in the transcript and not of the instrument. In Section 5, over-weighting is seen with the same models, but with different channel sets. The objection that our roster is too small to produce it becomes the observation that our instrument sees it only where the transcript enumerates the absences.

5 Example 2: identification or interpretation?

When an observer fails to reason on the absences, it may be failing to notice them or failing to understand what they mean in context. In general, these two are confounded, because a task that makes absences easier to identify usually also makes them easier to interpret.

Table 1 separates them without a new task. Its three shared-target conditions (FULL, ARITH, and RULED) have strictly nested operation sets, so moving up the ladder deletes exactly one operation and changes nothing else, and the deletion is done simply by changing the channels in the transcript. This does not have to be designed in; it falls out of the map.

To separate the failure to identify the absences from the failure to interpret them, we compare a configuration’s answers when the absences must be inferred (the derivable-absences condition) against its answers when they are listed out (the enumerated condition). Because both conditions state the same filing rule as seen in Appendix C, the only additional operation required with derivable-absences is to identify the $N - k$ absent reports from the stated analyst count. Whatever accuracy enumerating the absences recovers is what the configuration could not identify on its own.

However, that comparison only means something for a configuration that has trouble reading silence in the first place. Some configurations do not: they handle the condition where the absences must be inferred about as well as the condition where every result is printed, so they carry minimal difficulty in identifying and interpreting silence. Trying to quantify how much the enumeration of absences helps these configurations reason would be measuring repair where there was nothing to repair.

We separate these configurations with a written decision criterion. A configuration’s difficulty with silence is the gap between its error when the absences must be inferred and its error when every result is printed, $s_{1,\text{RULED}} - s_{1,\text{FULL}}$. A configuration for which that gap is below 0.15 levels is declared *at ceiling* and set aside as uninformative for identification localization. These configurations are at the ceiling of silence-reading ability, and this section asks about configurations with a genuine deficit in identifying absences. The threshold falls in an empty gap (Table 2).

Table 2: Silence-reading difficulty per admitted configuration: cell-weighted mean s_1 on the derivable-absences and printed-negatives conditions and their difference, over cells. The 0.15-level threshold falls in an empty gap. *sol-high* and *glm-high* sit well below it and near zero (*sol-high* is negative, scoring better when the absences must be inferred than when they are printed), so these two read silence with essentially no difficulty; the other three have a visible deficit, and the localization comparison is only valid for them.

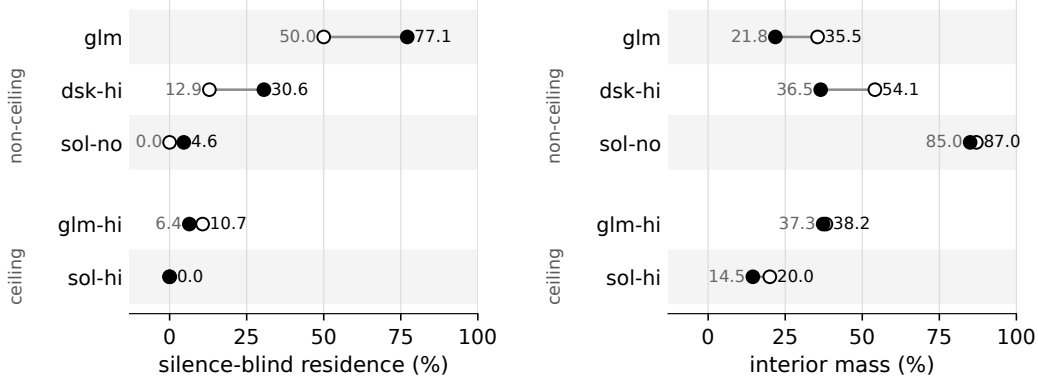
configuration	$s_{1,\text{FULL}}$	$s_{1,\text{RULED}}$	$s_{1,\text{RULED}} - s_{1,\text{FULL}}$
<i>sol-none</i>	0.431	0.762	0.331
<i>deepseek-high</i>	0.758	1.020	0.261
<i>glm</i>	0.764	2.243	1.479
<i>sol-high*</i>	0.200	0.114	−0.086
<i>glm-high*</i>	0.457	0.486	0.029

* at ceiling ($s_{1,\text{RULED}} - s_{1,\text{FULL}} < 0.15$ levels).

Identification is repaired. Deleting the identification step frees answers from the silence-blind belief. Moving from the derivable-absences condition to the enumerated one, occupancy of that belief collapses in every deficit configuration, from 4.6 to 0.0, from 30.6 to 12.9, and from 77.1 to 50.0 percent of draws (Figure 2a): the absences were going unnoticed, so enumerating them helped the configuration realize there were silences in the first place. We report that for the two ceiling configurations, *glm-high* moves the other way, 6.4 to 10.7, and *sol-high* has no occupancy to lose. Neither follows the pattern from the deficit configurations, as they have no genuine deficit to repair in the first place.

Interpretation is not. The answers freed from the silence-blind belief do not arrive at the target. Interior mass, the mass strictly between the silence-blind belief and the target posterior, rises in every deficit configuration as the step comes out, from 85.0 to 87.0, 36.5 to 54.1, and 21.8 to 35.5 percent on the cells that can express it, with D1–D3 dropped since these cells have the target and the silence-blind belief adjacent to each other, so no level lies between them (Figure 2b). Interior mass

rises for all five admitted configurations, but the two ceiling configurations are set aside since we established that they are uninformative for understanding the split in silence-reading.



(a) Occupancy of the silence-blind belief, over cells. It collapses for the three deficit configurations; the two ceiling configurations do not follow, with `glm-hi` rising instead. (b) Interior mass, on the cells that can express it (D1–D3 excluded, where target and silence-blind belief are adjacent). It rises for all five, but the two ceiling configurations are set aside.

Figure 2: Identification is repaired; interpretation is not. Moving from the derivable-absences condition (RULED) to the enumerated one (ARITH); filled marker RULED, open marker ARITH, rows grouped by the ceiling guard. (a) occupancy of the silence-blind belief collapses, so the absences were going unnoticed; (b) the freed mass rises into the interior rather than reaching the target (`dsk=deepseek`).

Identification is the one operation the derivable-absences condition needs that the enumerated one does not, so the collapse in occupancy of the silence-blind belief is evidence that enumerating the absences for the evaluated configuration improves its ability to recognize that the silences hold some evidential significance. However, the increase in interior mass is evidence that identifying the absences for the configuration is not enough to push it all the way to the correct answer.

The two halves meet here. Once the absences are enumerated, over-reading appears: two over-reads in 150 expressible admitted draws on the enumerated condition, against none in 300 on the two conditions that do not enumerate, and the two configurations producing them are exactly the two with the largest repairs. So the appearance of over-reading silence is a property of the channel set.

This localization is also recovered by a different method in Min et al. [2026]. Decomposing the closure judgment with structured certificates, they find that reasoning most often first diverges in how the model understands what the evidence covers. We reach a compatible split by removing required operations to see that while enumeration repairs identification, the belief still lands short of the target. The residual failure is interpretive in both cases. They hold the question and observed facts fixed and vary how much the evidence covers what the question asks, asking whether an absence justifies a negative answer at all, while we hold the target fixed and vary which operations the transcript requires to read the absences, asking how far belief should move.

6 Limitations and future directions

The instrument as designed does have its limitations:

- **Isolation against expressibility.** Coarsening each answer to one of four ordinal levels is what lets us read the direction and magnitude of a model’s update without confounding it with its ability to do Bayesian arithmetic, but that same coarseness leaves only a small, finite set of $(\ell_{\text{pos}}, \ell_{\text{tar}})$ configurations a cell can express, a limit the continuous framework does not have. Over-weighting is only expressible when the target posterior is not at an endpoint, which is only three out of fourteen cells. This is why the direction test of Figure 1 rests entirely on just D1–D3, and in those same cells, the target and the silence-blind belief

occupy adjacent levels, leaving no level between them, which is why D1–D3 are dropped from the interior-mass panel, Figure 2b. The two pull against each other: a coarser scale isolates the reasoning signal more cleanly but expresses fewer configurations, and a finer one expresses more but lets the arithmetic confound the silence-reading we are trying to measure.

- **Over-read against under-read visibility.** For cells D1–D3, $\ell_{\text{tar}} = 1$, which sits in $[0.40, 0.45]$, the narrowest band in the instrument. It has a gap of 0.20 below it and 0.10 above, so registering an over-read takes roughly twice the excursion an under-read does.
- **Resolution against candidate count.** With two candidates, the silence-blind belief is at least 0.5 by construction, so it will only ever occupy the top two levels. In the cells carrying Section 4’s result, the intermediate region is empty outright, so the 47–0 test lives on a one-level span, which follows from fixing $|A| = 2$.

Each of these limitations is also a direction for improving the instrument. The first and third share a repair, a richer coordinate system built off the existing framework: adding ordinal levels or moving beyond two candidates would increase the number of belief configurations the current scale cannot structurally express and make it easier to design cells for testing the way the observer can reside relative to the reference beliefs. The open question is how far the scale can be refined before the arithmetic confound that the coarse scale excluded returns. The second calls for a narrower fix: bands placed symmetrically about the target posterior would let an over-read and an under-read of the same size register equally, removing the advantage the scale currently gives to under-reading. None of these requires leaving the closed-form setting that makes the reference beliefs computable to begin with.

Reproduction. Stimuli, draw records, and analysis scripts are available at <https://github.com/akhan531/Control-On-LLMs>, commit 9ade4d4.

Acknowledgments

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A The cell design

Each row of Table 3 is a *cell*: one scenario put to the observers, fixed by an analyst count N , the number of filed positives k , and a per-assay strength p . The conditions of Table 1 are run on each of these fourteen cells, and the draws they return are the data behind every result in the paper. In the table, N , k , and p are the inputs; every other column is computed from them: the absences $N - k$, the silence-blind belief b_{pos} and the correct posterior b_{tar} , and the levels ℓ_{pos} and ℓ_{tar} that those two beliefs occupy.

Table 3: The fourteen cells. N , k , and p are inputs; the remaining columns are computed from them. b_{pos} is the belief that ignores the absences and b_{tar} the correct posterior, with ℓ_{pos} and ℓ_{tar} their ordinal levels. Beliefs are shown to three places.

cell	N	k	$N - k$	p	b_{pos}	b_{tar}	ℓ_{pos}	ℓ_{tar}
A1	5	2	3	0.81	0.948	0.190	3	0
A2	8	3	5	0.67	0.893	0.195	3	0
A3	10	4	6	0.67	0.944	0.195	3	0
A4	13	5	8	0.62	0.920	0.187	3	0
A5	14	5	9	0.67	0.972	0.056	3	0
B1	11	3	8	0.57	0.700	0.196	2	0
B2	5	1	4	0.62	0.620	0.187	2	0
B3	8	2	6	0.59	0.674	0.189	2	0
C1	6	2	4	0.88	0.982	0.018	3	0
C2	8	3	5	0.79	0.982	0.066	3	0
C3	11	4	7	0.73	0.982	0.048	3	0
D1	5	2	3	0.56	0.618	0.440	2	1
D2	8	3	5	0.54	0.618	0.421	2	1
D3	10	4	6	0.53	0.618	0.440	2	1

The cells fall into four blocks. **A** is the span-three backbone: the silence-blind belief sits at the top level ($\ell_{\text{pos}} = 3$) while the correct answer sits at the bottom ($\ell_{\text{tar}} = 0$), the widest displacement the scale allows, with N and k varied so that a level-three silence-blind belief is built from different amounts of evidence. **B** is the span-two backbone, the same construction one level before, with the silence-blind belief at level two; together A and B measure under-weighting at the two levels the silence-blind belief can occupy. **C** is a count control at level three: b_{pos} is held essentially constant (0.982 across the block) while k varies, so a configuration that reasons about the evidence answers the same across C1–C3 whereas one that merely tracks the number of filed positives does not, which separates competence from count-following. **D** is a count control at level two, and the one block built with $\ell_{\text{tar}} = 1$ by design: holding b_{pos} fixed at 0.618 plays the same control role as C, but placing the correct answer one level off the floor is what makes over-weighting expressible. D is the only block where an answer can land above the target as well as below it, and we built it that way on purpose so that under-weighting can be tested against over-weighting rather than only measured.

B The calibration pre-test

An observer responds with a candidate and confidence, both binary, producing the four ordinal levels from Section 3. The probability ranges those levels occupy are not assumed; the boundaries between the four levels are measured by a separate pre-test that hands each configuration a stated probability and records how the LLM maps that numeric value onto the ordinal levels. The pre-test is separate from the reported campaign: its records use the assay strength $p = 0.60$, which the main grid never uses (Appendix A). It runs the same seven configurations (Appendix D) over 32 stimuli, 16 in each of two families below, at three draws each (seeds 1–3); 659 of the 672 draws are usable. As everywhere in the paper there is no system prompt, decoding is strict against the schema below, and the probability bands are never shown to the model.

The response is a strict JSON object with two enumerated fields; the confidence field carries the two terms of whichever wording is in force, and the candidate field is one of the two printed names:

```
{
  "type": "json_schema",
  "json_schema": {
    "name": "silence_probe_ordinal",
    "strict": true,
    "schema": {
      "type": "object",
      "additionalProperties": false,
      "required": ["candidate", "confidence"],
      "properties": {
        "candidate": {
          "type": "string",
          "enum": ["SIGMA", "THETA"]
        },
        "confidence": {
          "type": "string",
          "enum": ["Somewhat confident", "Very confident"]
        }
      }
    }
  }
}
```

Two confidence wordings were tested: wording A ("Slightly"/"Strongly") and wording B ("Somewhat confident"/"Very confident"). The campaign and the map of Figure 3 use wording B, so the enum above shows its two terms.

The calibration sweep. Each stimulus states a probability outright, and the LLM observer has no record to read or inference to perform. One rendered stimulus (wording B, a stated 70% for the first candidate):

One of the following is responsible for the contamination event. Exactly one is.

Candidate SIGMA
Candidate THETA

An independent review has concluded that the chance SIGMA is responsible is 70 percent, and the chance THETA is responsible is 30 percent.

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".
"candidate" must be exactly SIGMA or THETA.
"confidence" must be exactly Somewhat confident or Very confident.

The stated probability is swept over $\{0.10, 0.20, 0.30, 0.40, 0.60, 0.70, 0.80, 0.90\}$ in steps of 0.10, with 0.50 omitted, since there is no favored candidate there. The sweep is mirrored, as 0.10/0.90, 0.20/0.80, 0.30/0.70 and 0.40/0.60 are mirror pairs, so the low edge and the high edge are measured separately rather than one assumed from the other. Taking each configuration’s modal answer across its three draws, the outermost stated probability at which the model still answered "Very confident" marks the switch edge. Under wording B, six of the seven configurations switch at 0.20 and 0.80, while llama is the exception, holding "Very confident" one step further in on the low side to 0.30. Because the sweep samples in steps of 0.10, the measured low edge at 0.20 and a nominal cut at 0.25 fall in the same step and cannot be told apart, so the band assignment agreeing under either is a resolution fact, not a robustness result.

The response bands. The same modal answers fix the whole four-level scale, not only the outer switch. Keeping the intervals where all seven configurations agree with the nominal 0.25/0.50/0.75 cut, and leaving the rest as gaps, yields these bands the coordinate system scores against:

level	answer	$P(\text{target})$ band
3	Very confident / target	[0.80, 1.00]
2	Somewhat confident / target	[0.55, 0.70]
1	Somewhat confident / other	[0.40, 0.45]
0	Very confident / other	[0.00, 0.20]

The three gaps are exactly where the sweep is not unanimous. The low gap runs from 0.20 to 0.40 rather than 0.30 because llama answers "Very confident" at 0.30 where the other six answer "Somewhat confident", so no level is agreed there. The 0.70–0.80 gap is the matching high-side transition. The 0.45–0.55 gap is deliberately placed around the 0.50 prior, where the favored candidate would flip. Every predicted belief in the grid (Appendix A) is built to land inside a band, so no cell’s correct level depends on where a model’s personal threshold sits.

The format decision. A second family checks that the confidence field actually moves with the belief. Each stimulus is a positives-only record of k POSITIVE reports ($k \in \{1, 2, 4, 6\}$, with $k = 1$

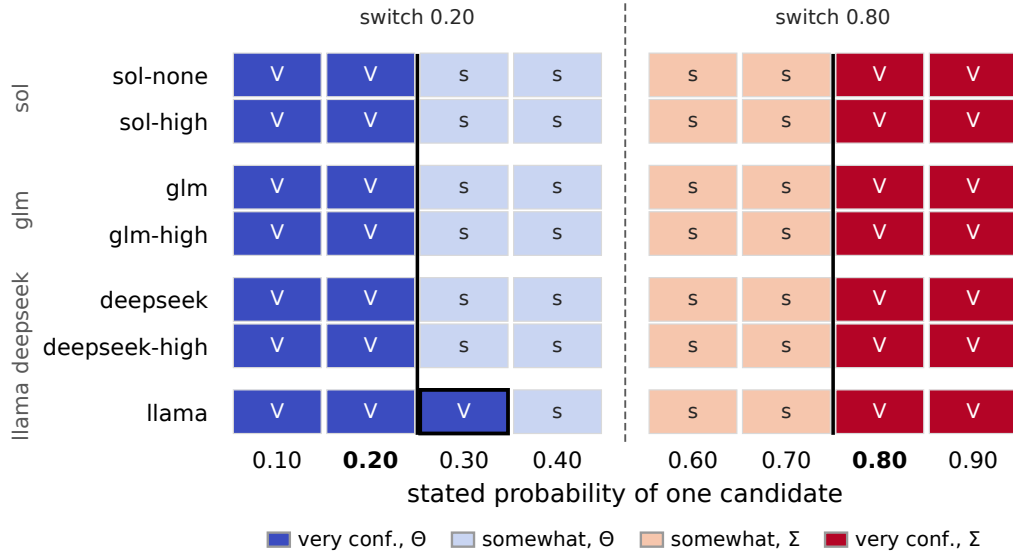


Figure 3: The calibration confidence map. Each row is one configuration; reading its cells outward from even odds, the answer switches from somewhat to very confident. Six of the seven switch together at the 0.20 and 0.80 ticks (bold); llama is the exception, holding very confident one step further in, to 0.30 on the low side (boxed). Color encodes the answer given most often in each cell: hue is the chosen candidate, Σ or Θ , and shade its confidence, dark for very and pale for somewhat.

and $k = 6$ the decision pair) at assay strength 0.60, stating neither the analyst count nor the filing rule, so the belief rises with k while nothing in the text names the operation. One rendered stimulus (wording B, $k = 6$):

One of the following is responsible for the contamination event. Exactly one is.

Candidate SIGMA
Candidate THETA

Before any testing, each candidate is considered equally likely.

Analysts were each assigned a different assay. Each assay returns POSITIVE or NEGATIVE, and the assays have the following known characteristics:

If SIGMA is responsible, an assay returns POSITIVE with probability 0.60.
If THETA is responsible, an assay returns POSITIVE with probability 0.40.

Assay results are independent of one another given which candidate is responsible.

REPORTS ON FILE

Assay returned POSITIVE
Assay returned POSITIVE
Assay returned POSITIVE
Assay returned POSITIVE
Assay returned POSITIVE
Assay returned POSITIVE

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".
"candidate" must be exactly SIGMA or THETA.
"confidence" must be exactly Somewhat confident or Very confident.

All seven configurations moved the confidence term in the intended direction, answering "Somewhat confident" at $k = 1$ and "Very confident" at $k = 6$, which solidified the four-level scale.

C Prompts and elicitation

There is no system prompt. Each draw is a single user message carrying the entire prompt below; no separate system or developer message is sent. The model returns a structured object validated against this schema, with strict decoding and no partial credit:

```
{
  "type": "json_schema",
  "json_schema": {
    "name": "silence_probe_v2_ordinal",
    "strict": true,
    "schema": {
      "type": "object",
      "properties": {
        "candidate": {
          "type": "string",
          "enum": ["SIGMA", "THETA"]
        },
        "confidence": {
          "type": "string",
          "enum": ["Somewhat confident", "Very confident"]
        }
      },
      "required": ["candidate", "confidence"],
      "additionalProperties": false
    }
  }
}
```

The two enumerated answer fields are therefore candidate, allowed values SIGMA and THETA, and confidence, allowed values Somewhat confident and Very confident. Together they give the four ordinal levels of Section 3.

The four conditions of Table 1 are generated by two scripts: `silence_v2_stimuli.py` (FULL, BLIND, RULED) and `arith_d0_stimuli.py` (ARITH). Both import and assert the same forbidden-word list, so the no-silence-vocabulary property holds across every condition, not a subset. The list has nineteen terms:

```
silent, silence, absent, absence, missing, did not report, no report,
nothing was filed, unreported, withheld, omitted, posterior, bayes,
prior probability, likelihood ratio, probability that, update your,
calculate, compute
```

Below is one fully rendered stimulus per condition, all from cell A1 ($N = 5$, $k = 2$, positive rates 0.81 and 0.19), first label ordering. Only the middle block changes across conditions; the preamble, the assay description, and the elicitation are identical.

FULL.

One of the following is responsible for the contamination event. Exactly one is.

```
Candidate SIGMA
Candidate THETA
```

Before any testing, each candidate is considered equally likely.

5 analysts were each assigned a different assay. Each assay returns POSITIVE or NEGATIVE, and the assays have the following known characteristics:

```
If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
If THETA is responsible, an assay returns POSITIVE with probability 0.19.
```

Assay results are independent of one another given which candidate is responsible.

COMPLETE RESULTS, ALL 5 ASSAYS

```
Assay returned POSITIVE
Assay returned POSITIVE
Assay returned NEGATIVE
Assay returned NEGATIVE
Assay returned NEGATIVE
```

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".
"candidate" must be exactly SIGMA or THETA.
"confidence" must be exactly Somewhat confident or Very confident.

BLIND. Neither N nor the filing rule is stated.

One of the following is responsible for the contamination event. Exactly one is.

Candidate SIGMA
Candidate THETA

Before any testing, each candidate is considered equally likely.

Analysts were each assigned a different assay. Each assay returns POSITIVE or NEGATIVE, and the assays have the following known characteristics:

If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
If THETA is responsible, an assay returns POSITIVE with probability 0.19.

Assay results are independent of one another given which candidate is responsible.

REPORTS ON FILE

Assay returned POSITIVE
Assay returned POSITIVE

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".
"candidate" must be exactly SIGMA or THETA.
"confidence" must be exactly Somewhat confident or Very confident.

RULED. N and the filing rule are stated; the absences are derivable but not printed.

One of the following is responsible for the contamination event. Exactly one is.

Candidate SIGMA
Candidate THETA

Before any testing, each candidate is considered equally likely.

5 analysts were each assigned a different assay. Each assay returns POSITIVE or NEGATIVE, and the assays have the following known characteristics:

If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
If THETA is responsible, an assay returns POSITIVE with probability 0.19.

Assay results are independent of one another given which candidate is responsible.

Each analyst files a report only when their assay returns POSITIVE.

REPORTS ON FILE

Assay returned POSITIVE
Assay returned POSITIVE

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".
"candidate" must be exactly SIGMA or THETA.
"confidence" must be exactly Somewhat confident or Very confident.

ARITH. The absences are printed as filing lines.

One of the following is responsible for the contamination event. Exactly one is.

Candidate SIGMA
Candidate THETA

Before any testing, each candidate is considered equally likely.

5 analysts were each assigned a different assay. Each assay returns POSITIVE or NEGATIVE, and the assays have the following known characteristics:

If SIGMA is responsible, an assay returns POSITIVE with probability 0.81.
If THETA is responsible, an assay returns POSITIVE with probability 0.19.

Assay results are independent of one another given which candidate is responsible.

Each analyst files a report only when their assay returns POSITIVE.

COMPLETE FILING RECORD, ALL 5 ANALYSTS

Assay returned POSITIVE
Assay returned POSITIVE
Filed nothing at all
Filed nothing at all
Filed nothing at all

Each analyst appears exactly once above.

Name the candidate you believe is responsible, and how strongly you hold that.

Reply with only a JSON object with keys "candidate" and "confidence".

"candidate" must be exactly SIGMA or THETA.

"confidence" must be exactly Somewhat confident or Very confident.

Both label orderings. Every stimulus appears in two orderings; the mirror swaps which printed candidate carries which positive rate. The FULL stimulus above in its second ordering differs only in these two lines, so a fixed label bias is separable from a competent answer:

If SIGMA is responsible, an assay returns POSITIVE with probability 0.19.
If THETA is responsible, an assay returns POSITIVE with probability 0.81.

D Models and inference

Reasoning setting and token budget move together; the "-high" member of a pair raises both. The gate score is the draw-weighted mean s_1 on the printed-negatives condition (FULL), the competence gate of Section 3.

alias	model slug	reasoning	max_tokens	temp.	weights	gate s_1
sol-hi	openai/gpt-5.6-sol	effort high	12000	—	closed	0.200
sol-no	openai/gpt-5.6-sol	effort none	800	—	closed	0.421
glm-hi	z-ai/glm-5.2	enabled	16000	0.7	open	0.457
dsk-hi	deepseek/deepseek-v4-flash-0731	enabled	16000	0.7	open	0.752
glm	z-ai/glm-5.2	disabled	800	0.7	open	0.764
deepseek	deepseek/deepseek-v4-flash-0731	disabled	800	0.7	open	1.609
llama	meta-llama/llama-3.3-70b-instruct	none	400	0.7	open	1.800

Weight availability is the model provider's; the repository records the slugs above, not licenses (see Appendix E).

The five configurations above the rule are admitted by the gate; deepseek and llama are excluded. The openai/gpt-5.6-sol model does not accept a temperature, so both sol configurations vary by seed alone.

All calls went through the OpenRouter API. Each stimulus was drawn five times, one per seed (seeds 1 through 5); the campaign holds five draws per stimulus. The response was parsed strictly against the schema in Appendix C with no partial credit, and each draw was classified into one of four outcomes: TRUNCATED (the completion hit the token budget), PARSE (unparseable or off-menu), REQUEST (transport or HTTP failure), and AuthError (a 401/403, which stops the run immediately). Transient failures were retried up to three times with exponential backoff; authentication failures

were not retried. The run resumes by keying on successful draws only, so a failed draw is re-attempted rather than skipped. Token budgets are per configuration, and reasoning tokens are drawn from the same budget, which is why the "-high" configurations carry the larger `max_tokens`.

No claim in Sections 4, 5, or 6 compares two configurations that share a model slug; every comparison is across conditions or across the roster, so the reasoning-and-budget difference within a pair does not enter any reported result.

E Data and licensing

The repository carries no LICENSE file, and per-model terms of use are not recorded in it: it records the model slugs of Appendix D but not their licenses. Model access was through the OpenRouter API. We state what the repository does and does not contain rather than reproducing terms it does not hold.

F Use of AI assistance

Coding agents and large language models assisted this project in two ways: executing and iterating the analysis pipeline in the repository, including data analysis and figure generation, and drafting and editing the manuscript. Every numeric quantity in the paper was verified independently of that assistance, recomputed from the draw records at the cited commit or, where the quantity is analytic, re-derived from the model's closed form; every citation was checked against its primary source; the design, the claims, and their scoping are the author's.